Computer Vision

Precision Livestock Farming

Instituto Federal de Goiás - IFG
Artificial Intelligence

Prof. Dr. Eduardo Noronha

Agenda

Keypoints (Pose Estimation)

Applications

Annotation

Model inference

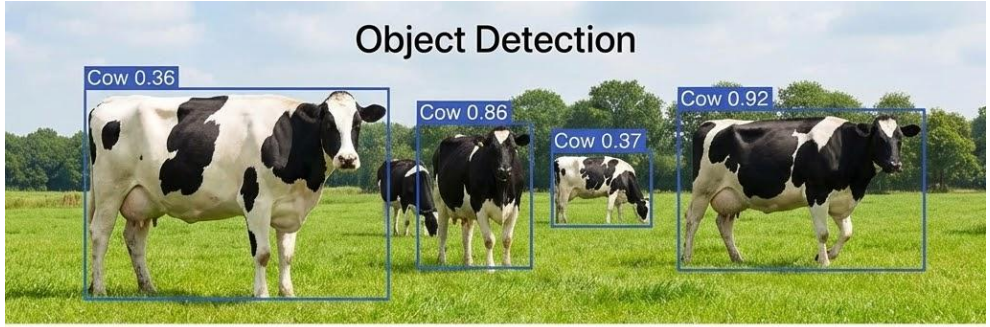
Metrics

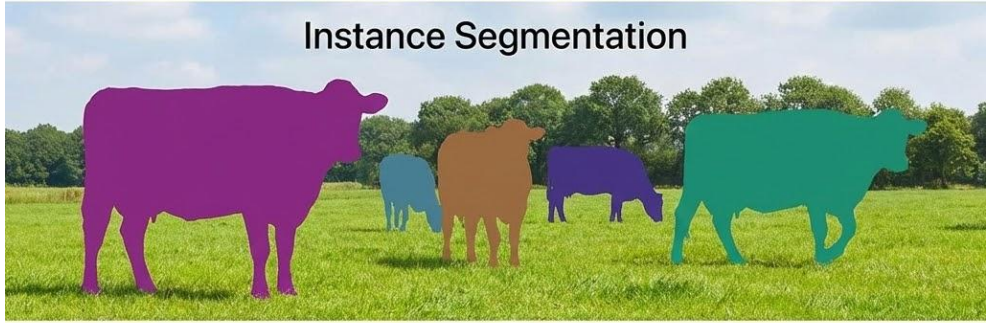
Training

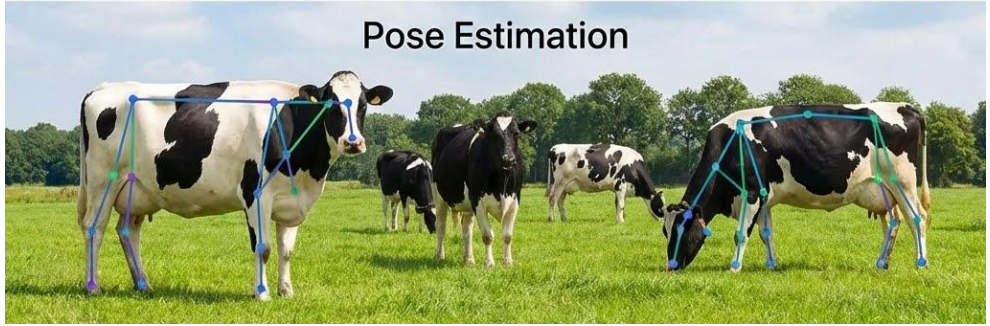
Case study on Dairy

- Keypoints detection
- Metrics for ML
- Results

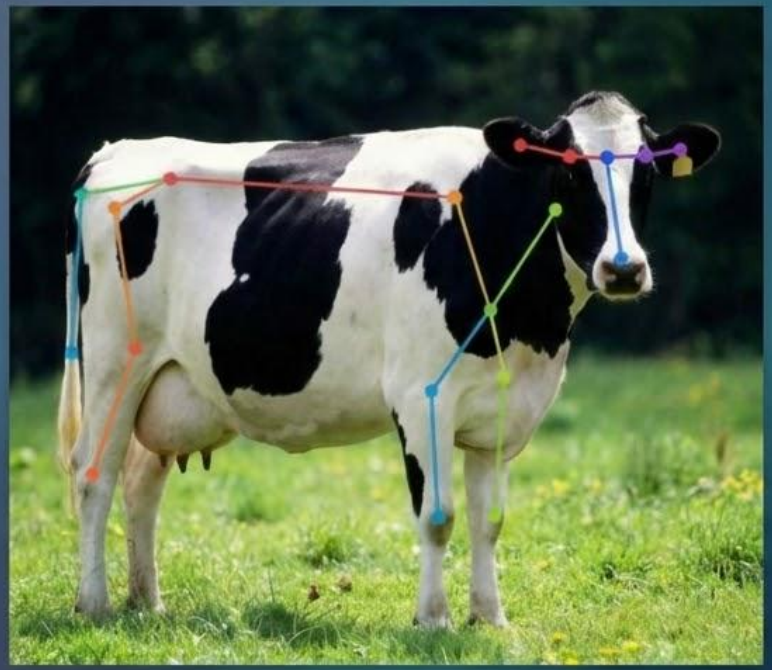
Industry Challenge







Pose Estimation



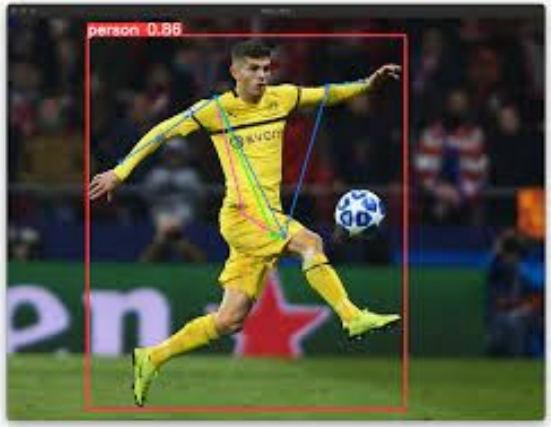
Goal:

Involves identifying the location of specific points in an image (keypoints, 2D \[x, y]] or 3D \[x, y, visible]]).

Output:

Set of points that represent the **keypoints** on an object in the image, usually along with the confidence scores for each point.

Applications



Annotation

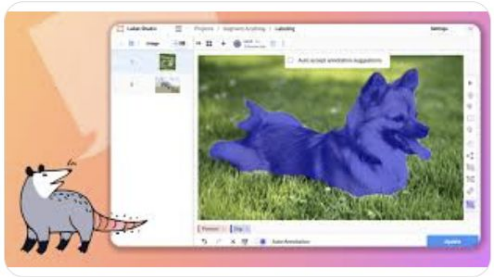
(0,0)

(100,100)

```
docker run -it -p 8080:8080 \
    -v $(pwd)/mydata:/label-studio/data \
    --env LABEL_STUDIO_LOCAL_FILES_SERVING_ENABLED=true \
    --env LABEL_STUDIO_LOCAL_FILES_DOCUMENT_ROOT=/label-studio/data/images \
    heartexlabs/label-studio:latest
```

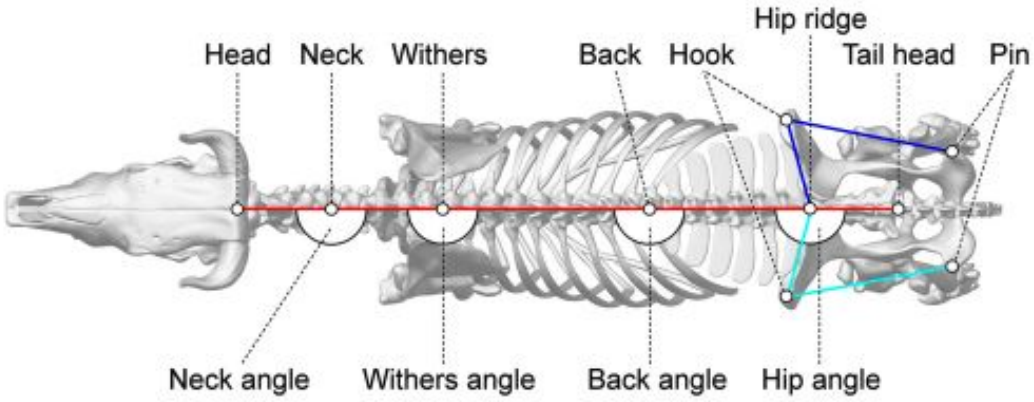
Label Studio
by HumanSignal

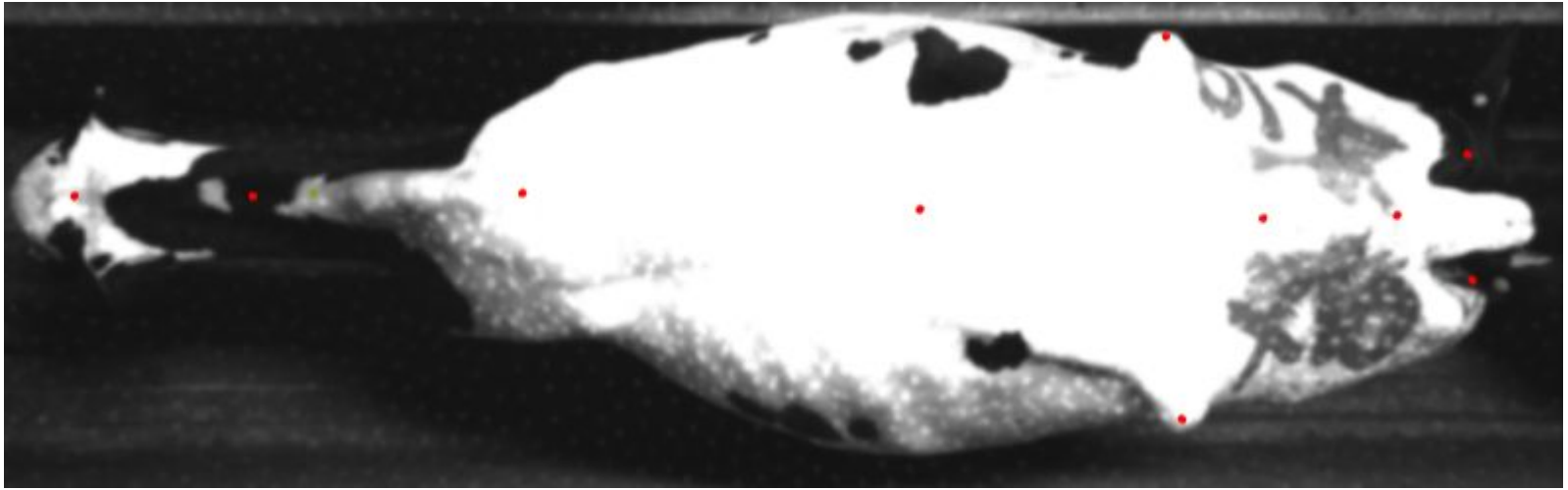
Getting started usin...
labelstud.io/blog/get-star...



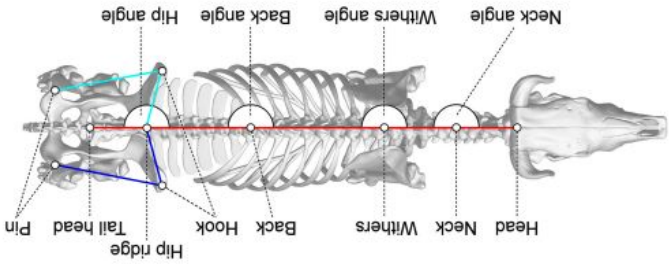
Install

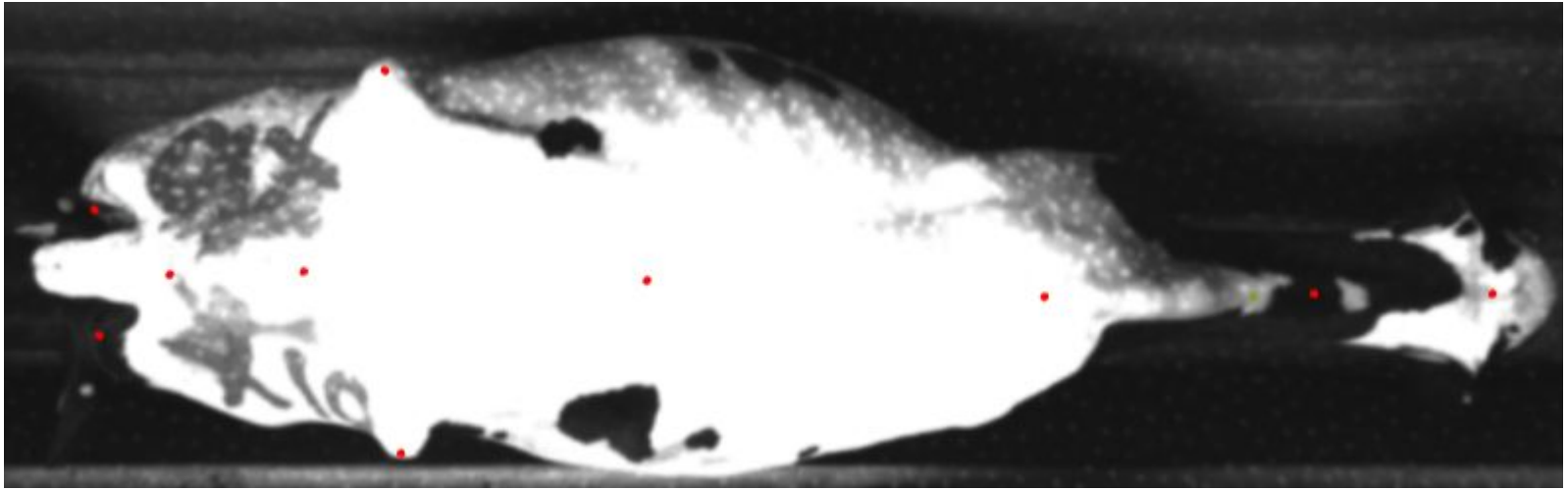
Annotation


Hip ridge
Head
Neck
Withers
Back
Hook
Tail head
Pin
Neck angle
Withers angle
Back angle
Hip angle

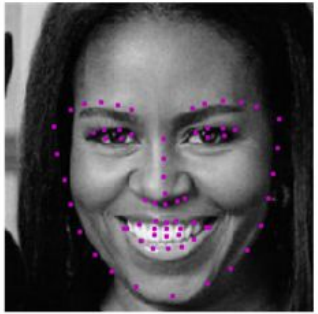


Annotation


Head
Neck
Withers
Back
Hook
Hip ridge
Tail head
Pin
Neck angle
Withers angle
Back angle
Hip angle



YOLO Dataset Format



1. One text file per image: Each image in the dataset has a corresponding text file with the same name as the image file and the ".txt" extension.
2. One row per object: Each row in the text file corresponds to one object instance in the image.
3. Object information per row: Each row contains the following information about th
   - Object class index: An integer representing the class of the object (e.g., 0 for
   - Object center coordinates: The x and y coordinates of the center of the objec between 0 and 1.
   - Object width and height: The width and height of the object, normalized to be between 0 and 1.
   - Object keypoint coordinates: The keypoints of the object, normalized to be between 0 and 1.

- 0 : Not labeled
- 1 : Labelled but not visible
- 2 : Labeled and visible.

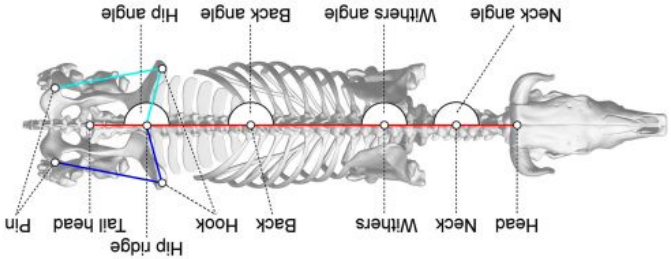
<class-index> <x> <y> <width> <height> <px1> <py1> <v1> <px2> <py2> <v2> ... <pxn> <pyn> <vn>

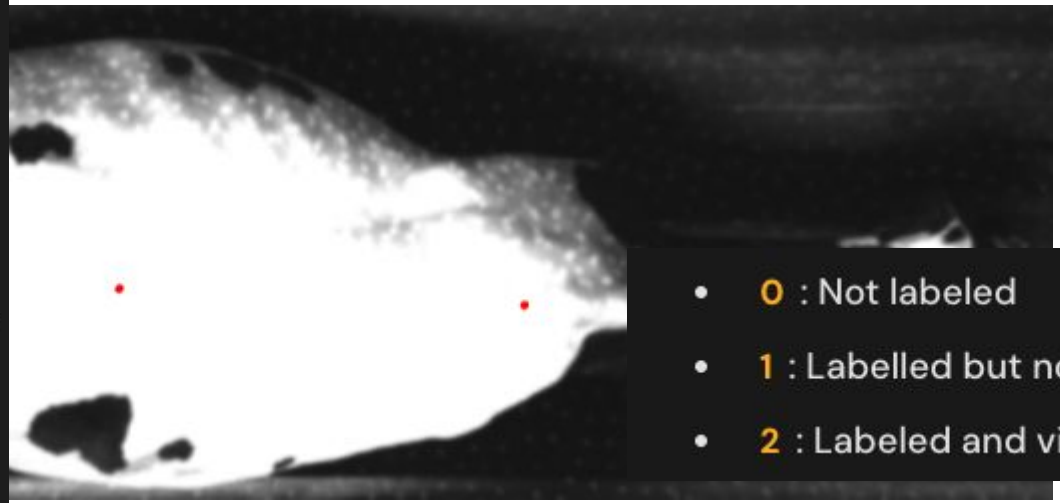
<class> Bounding box keypoint 1 keypoint 2 ... keypoint n

Annotation

Json format


```
{
    "id": 88,
    "annotations": [
        {
            "id": 3,
            "completed_by": 1,
            "result": [
                {
                    "original_width": 640,
                    "original_height": 480,
                    "image_rotation": 0,
                    "value": {
                        "x": 70.76053442959919,
                        "y": 58.27338129496403,
                        "width": 0.15416238437821173,
                        "keypointlabels": [
                            "neck"
                        ]
                    },
                    "id": "r4N0GZcOhy",
                    "from_name": "kp-1",
                    "to_name": "img-1",
                    "type": "keypointlabels",
                    "origin": "manual"
                },
```



- 0 : Not labeled
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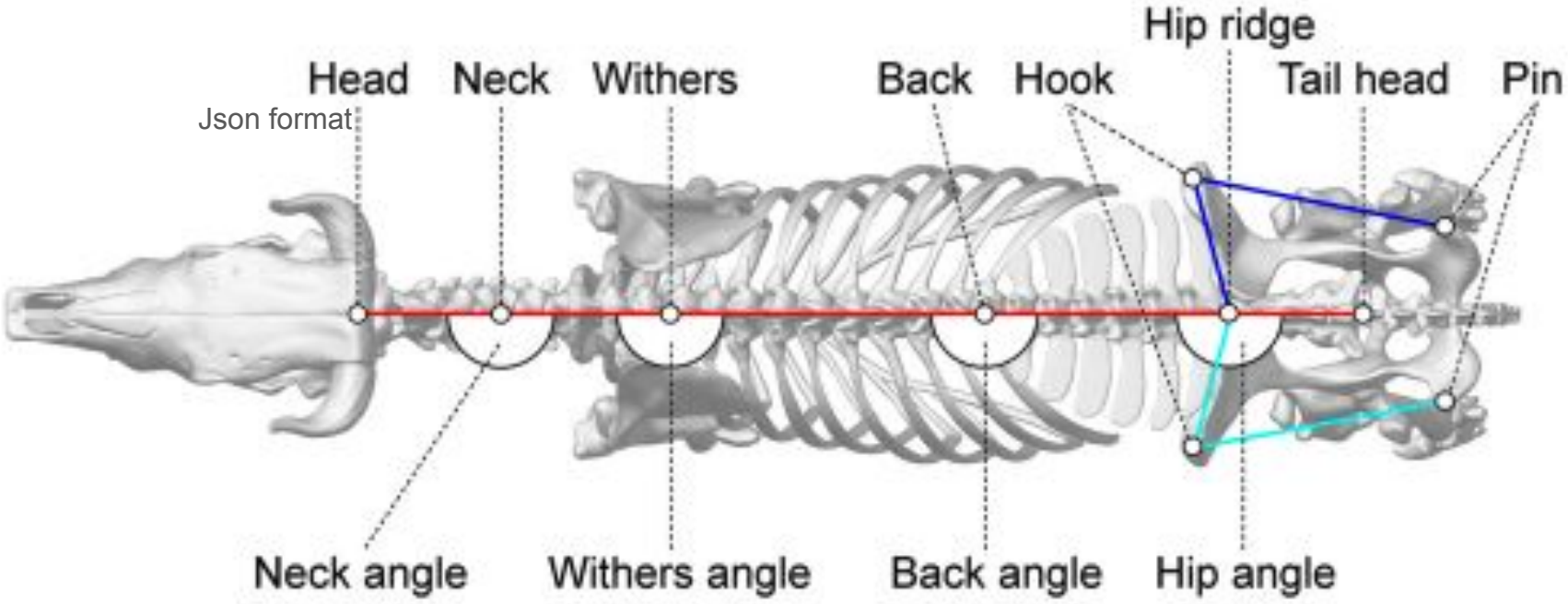
<class> Bounding box keypoint 1 keypoint 2 ... keypoint n

<class-index> <x> <y> <width> <height> <px1> <py1> <v1> <px2> <py2> <v2> ... <pxn> <pyn> <vn>

<class> Bounding box keypoint 1 keypoint 2 ... keypoint n



Json format

<class> Bounding box keypoint 1 keypoint 2 ... keypoint n

<class-index> <x> <y> <width> <height> <px1> <py1> <v1> <px2> <py2> <v2> ... <pxn> <pyn> <vn>

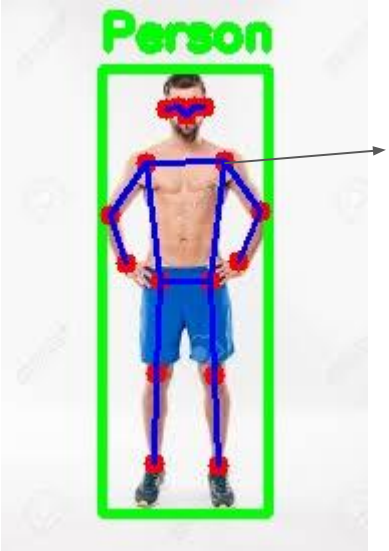
<class> Bounding box keypoint 1 keypoint 2 ... keypoint n

Pretrained Models

COCO dataset + ImageNet dataset.

Model	size (pixels)	mAP^{pose} 50-95(e2e)	mAP^{pose} 50(e2e)	Speed CPU ONNX (ms)	Speed T4 TensorRT10 (ms)	params (M)	FLOPs (B)
YOLO26n-pose	640	57.2	83.3	40.3 ± 0.5	1.8 ± 0.0	2.9	7.5
YOLO26s-pose	640	63.0	86.6	85.3 ± 0.9	2.7 ± 0.0	10.4	23.9
YOLO26m-pose	640	68.8	89.6	218.0 ± 1.5	5.0 ± 0.1	21.5	73.1
YOLO26l-pose	640	70.4	90.5	275.4 ± 2.4	6.5 ± 0.1	25.9	91.3
YOLO26x-pose	640	71.6	91.6	565.4 ± 3.0	12.2 ± 0.2	57.6	201.7

Dataset Format




```
5 0.6 0.5 0.2 0.7 0.48 0.45 1 0.52 0.45 1 0.5 0.48
1
```

```
tensor([[[ 92.6279,  57.3633],
         [ 96.2803,  53.0127],
         [ 88.4385,  53.1738],
         [101.1680,  53.8184],
         [ 82.4766,  54.3018],
         [111.6953,  80.3516],
         [ 72.3252,  81.6406],
         [129.2051, 105.4346],
         [ 54.5469, 107.0996],
         [118.8926, 130.5176],
         [ 62.6035, 131.4844],
         [104.3369, 140.2930],
         [ 79.6836, 140.4004],
         [106.4316, 187.1289],
         [ 77.3203, 186.2695],
         [108.3652, 232.0312],
         [ 76.0312, 231.8164]]]
```

<class-index> <x> <y> <width> <height> <px1> <py1> <px2> <py2> ... <pxn> <pyn>

Inference

```python
from ultralytics import YOLO

# Run inference on an image
results = pose_model(image_path)

# Process and visualize the results
for result in results:
    # Keypoints in normalized format can be accessed
    keypoints = result.keypoints.xy # XY coordinates
    conf = result.keypoints.conf    # Confidence score
    data = result.keypoints.data    # XY and confidence score together
    print(keypoints, conf, data)
```

Code Example

https://colab.research.google.com/drive/1m-3LgTd_AeMb6L9N1yIFZg8BRCxR76jO?usp=sharing

Evaluation metrics

$$KS_i = \exp\left(\frac{-d_i^2}{2s^2k_i^2}\right)$$

Keypoint similarity

- d_i is the Euclidean distance between the ground truth and predicted keypoint i
- k is the constant for keypoint i
- s is the scale of the ground truth object; s^2 hence becomes the object's segmented area.

$$OKS = \frac{\sum_i \exp\left(\frac{-d_i^2}{2s^2k_i^2}\right)\delta(v_i > 0)}{\sum_i \delta(v_i > 0)}$$

Object Keypoint Similarity (OKS)

- KS_i is the keypoint similarity for keypoint type i
- v_i is the ground truth visibility flag for keypoint i
- $\delta(v_i > 0)$ is the Dirac-delta function which computes as 1 if the keypoint i is labeled, otherwise 0

Keypoints	δ_i
Nose	0.026
Eyes	0.025
Ears	0.035
Shoulders	0.079
Elbows	0.072
Wrists	0.062
Hips	0.107
Knees	0.087
Ankles	0.089

k=2

Source: https://learnopencv.com/object-keypoint-similarity/

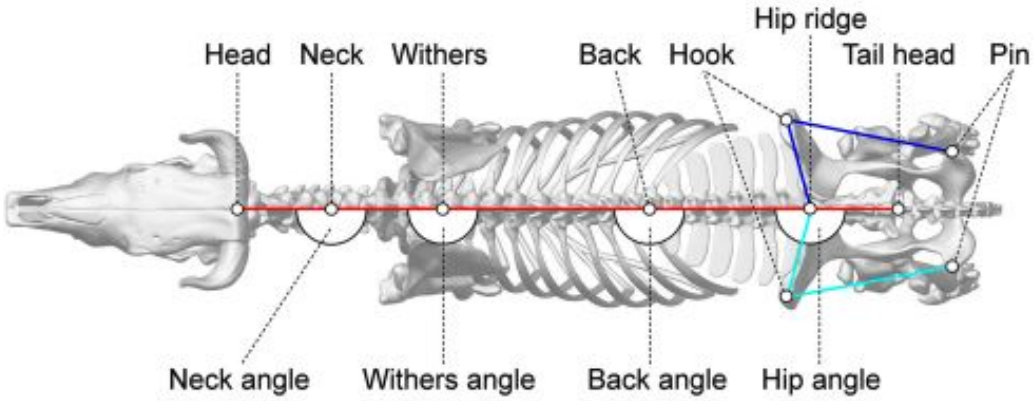
Training

```yaml
# Train/val/test sets as 1) dir: path/to/imgs, 2) file: path/to/imgs.txt, or 3) list: [path/
path: coco8-pose # dataset root dir
train: images/train # train images (relative to 'path') 4 images
val: images/val # val images (relative to 'path') 4 images
test: # test images (optional)

# Keypoints
kpt_shape: [17, 3] # number of keypoints, number of dims (2 for x,y or 3 for x,y,visible)
flip_idx: [0, 2, 1, 4, 3, 6, 5, 8, 7, 10, 9, 12, 11, 14, 13, 16, 15]

# Classes
names:
  0: person

# Keypoint names per class
kpt_names:
  0:
    - nose
    - left_eye
    - right_eye
    - left_ear
    - right_ear
    - left_shoulder
    - right_shoulder
    - left_elbow
```

coco8-pose.yaml

Example source: https://learnopencv.com/animal-pose-estimation/

Challenge



Objective dairy cow mobility analysis and scoring system using computer vision–based keypoint detection technique from top-view 2-dimensional videos, JDS, 2024, https://doi.org/10.3168/jds.2024-25545.

Dataset information
30 cows at the parlor milking station area
50 images for each cow
Files names pattern:

cow_id_YYYY_MM_DD_HH_MM_SS_cam_ID_satation_id.jpg

cam_ID_XX_YYYYMMDDHHMMSS_station_id_cam_ID.jpg
YYYYMMDD_HHMMSS_station_id_cam_ID.jpg

cow_id: number of the cow
YYYY: year
MM: month:
DD: day
HH hour
MM: minute
SS: second
cam_ID: number of the camera
station_ID: station position in the parlor

Required tasks

1. Annotate Keypoints Dataset for cows
2. Train and evaluate a Keypoint Detection Model
3. Generate features that could potentially identify each animal
4. Conduct a descriptive analysis of the features and its usability for identification
5. Design and train a machine learning model to classify the animals
6. Evaluate the model

https://drive.google.com/drive/folders/1i3JTjED8r_8qseQfNTQgOxSR5X68kpPD

Next Meeting

Industry Challenges